EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM

To create a 3Dmodel from the given isometric drawing using fusion 360 tool.

REQUIREMENTS

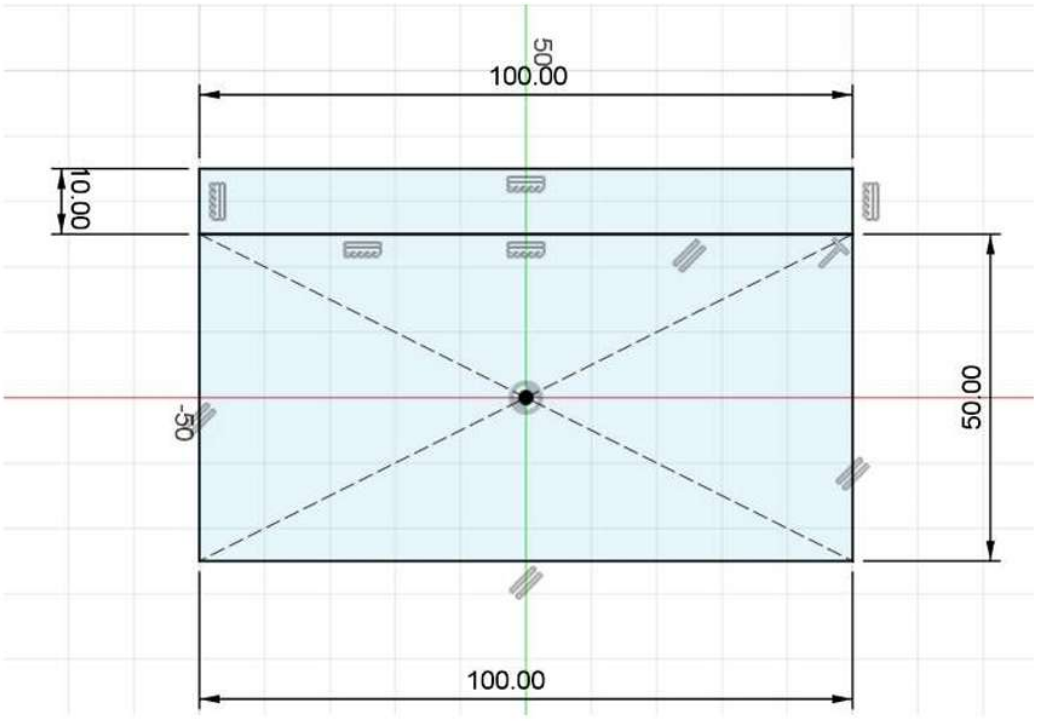
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE

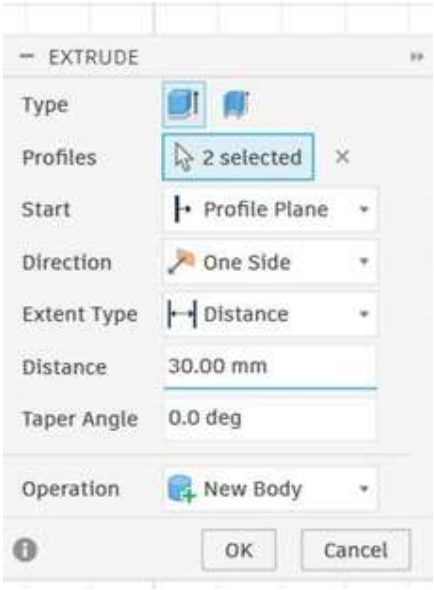
Step 1: Creating the Main Body

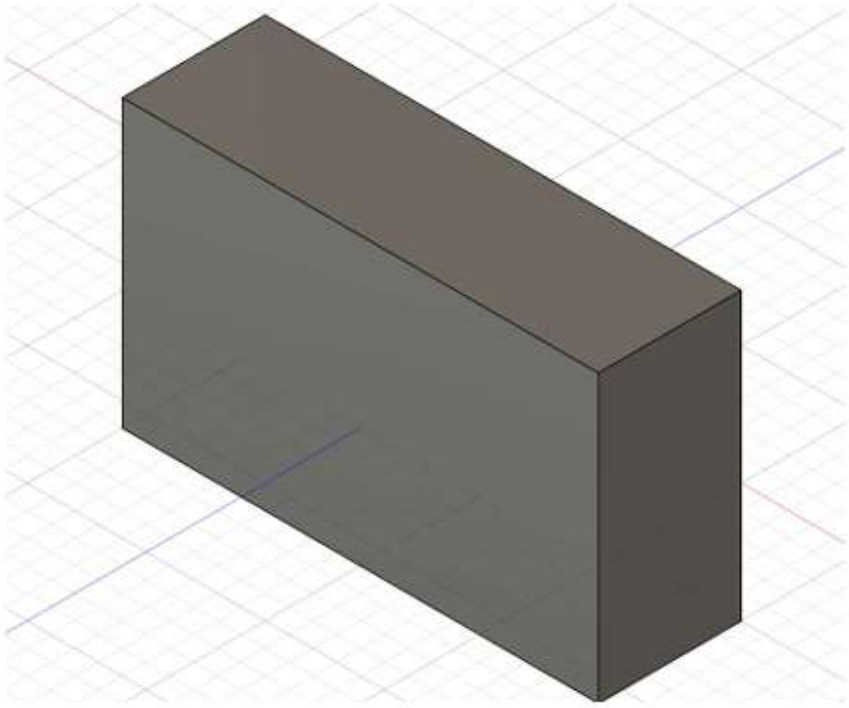
Objective: Create the primary rectangular housing of the camera.

1. Create Sketch: Click on Create Sketch and select the Front Plane (XY plane).
2. Draw Rectangle: Select the Two-Point Rectangle tool from the sketch palette. Click on the origin (0,0), drag outwards, and set dimensions:
    - Width: 100 mm
    - Height: 50 mm



3. Finish Sketch: Click Finish Sketch in the top right toolbar.
4. Extrude 1 Details:
    - Press E (or go to Create > Extrude).
    - Select the rectangular profile.
    - Direction: One-side.
    - Distance: 35 mm (Depth of the main camera body based on the top-view layout dimensions).
    - Operation: New Body.
    - Click OK.



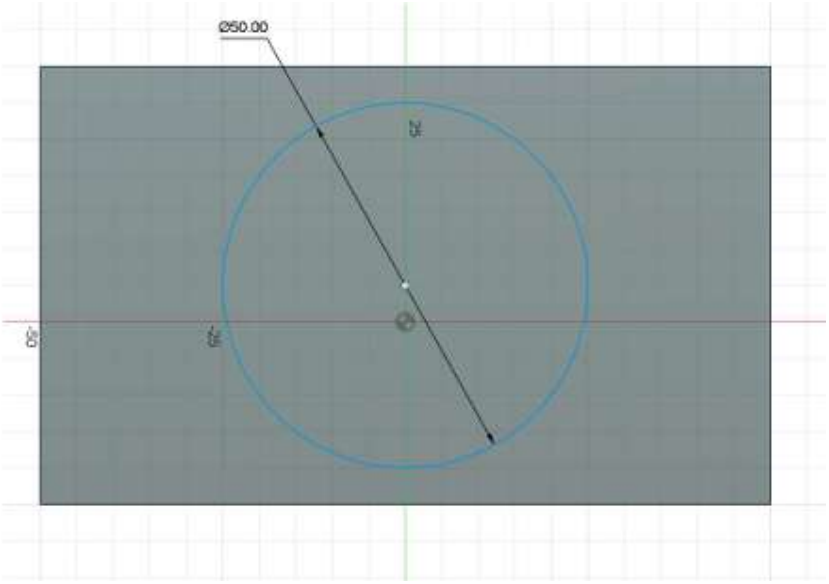


Step 2: Creating the Main Camera Lens

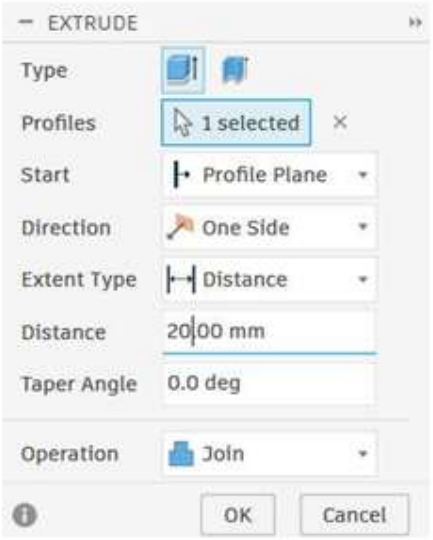
Objective: Model the prominent circular lens protruding from the front of the camera body.

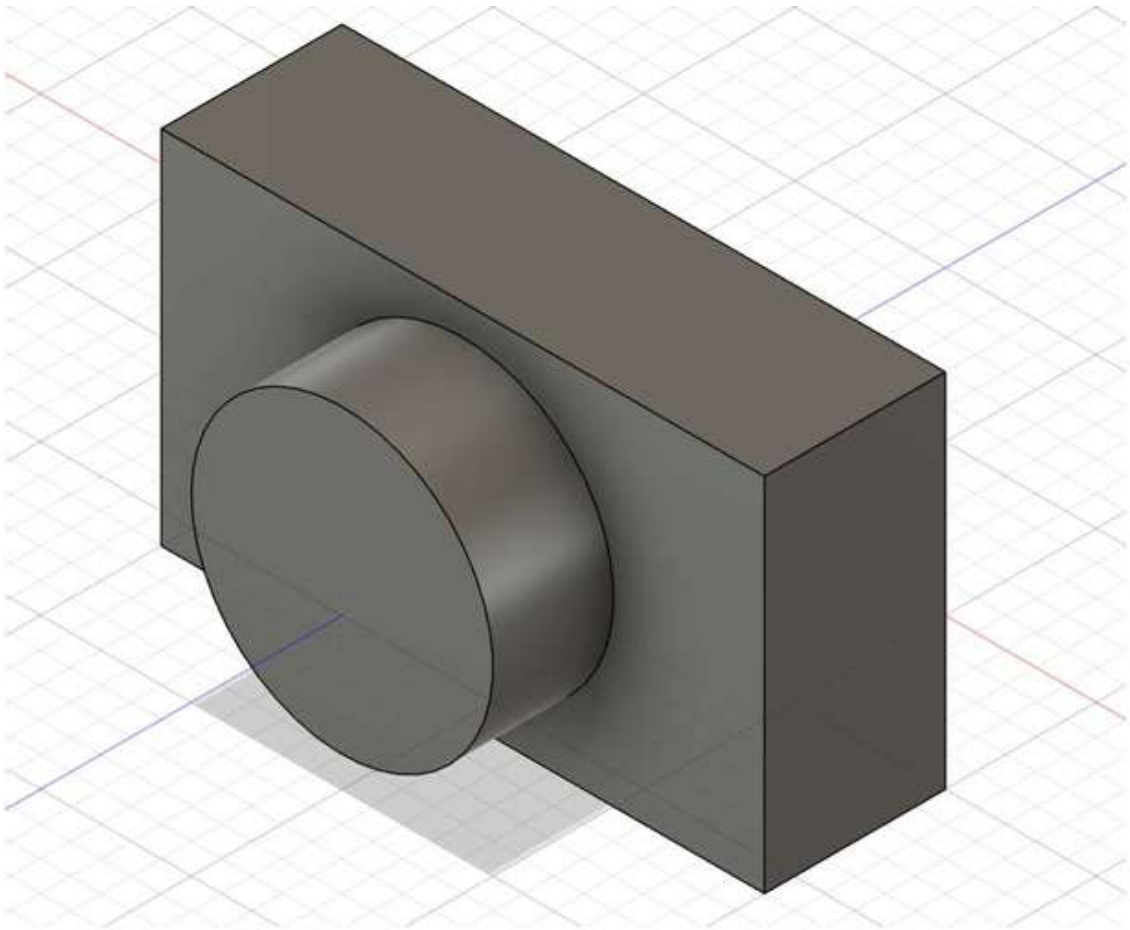
1. Create Sketch: Click Create Sketch and select the Front Face of the main camera body as your sketch plane.
2. Draw Circle: Select the Center Diameter Circle tool (C).

- Locate the center point by snapping to the vertical centerline of the front face.
- Draw a circle with a diameter of Ø50 mm.
- Add dimension constraints to position it centrally if necessary.



3. Finish Sketch: Click Finish Sketch.
4. Extrude 2 Details:
    - Press E to open the Extrude dialog box.
    - Select the circular profile of the lens.
    - Direction: One-side.
    - Distance: 20 mm (as specified for the lens protrusion depth in the drawing).
    - Operation: Join (so it merges smoothly with the main body).
    - Click OK.

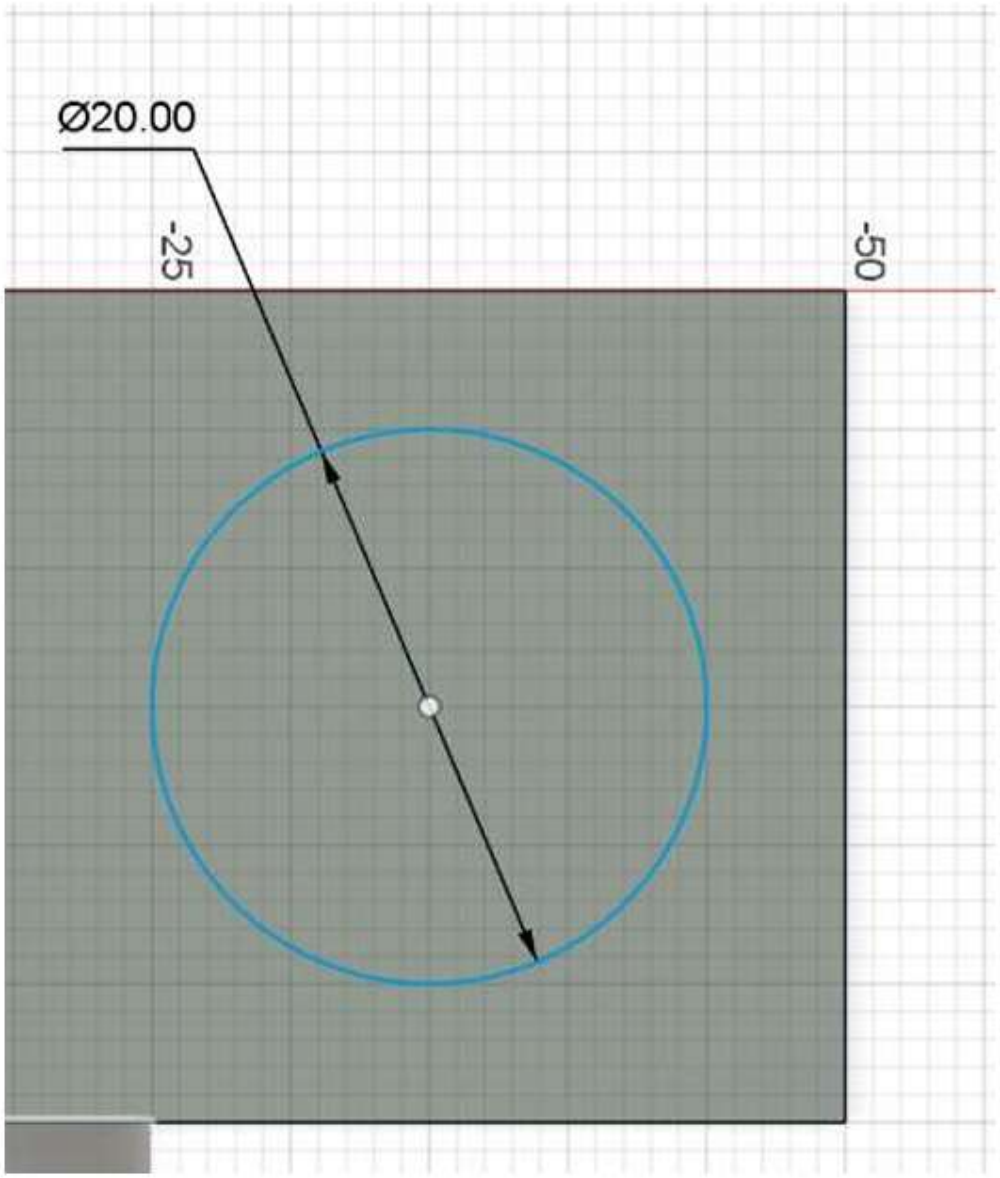

EXTRUDE
Type
Profiles
1 selected
Start
Profile Plane
Direction
One Side
Extent Type
Distance
Distance
20.00 mm
Taper Angle
0.0 deg
Operation
Join
OK
Cancel



Step 3: Creating the Top-Right Circular Button

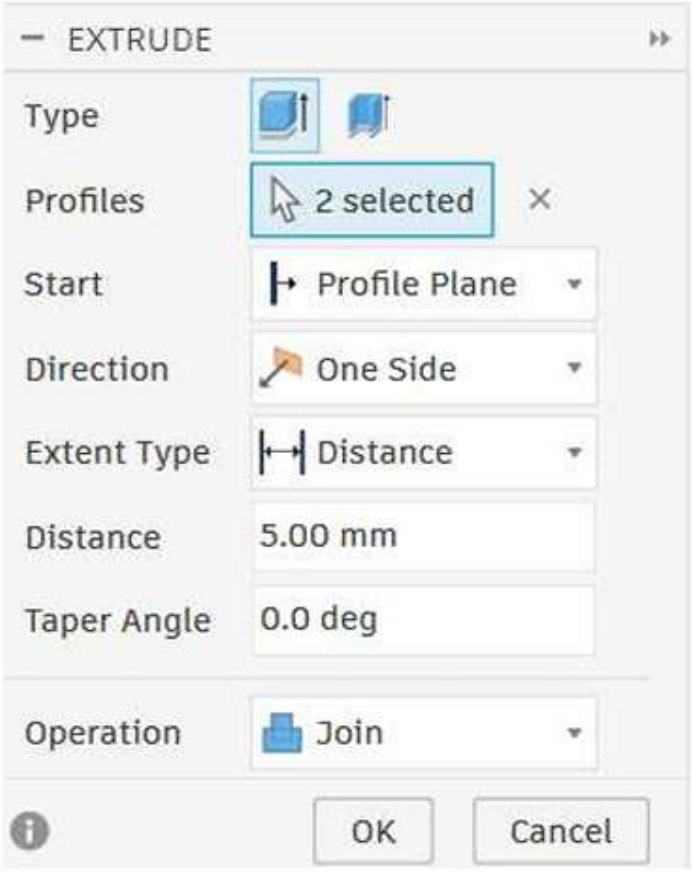
Objective: Model the cylindrical control button located on the top right surface.

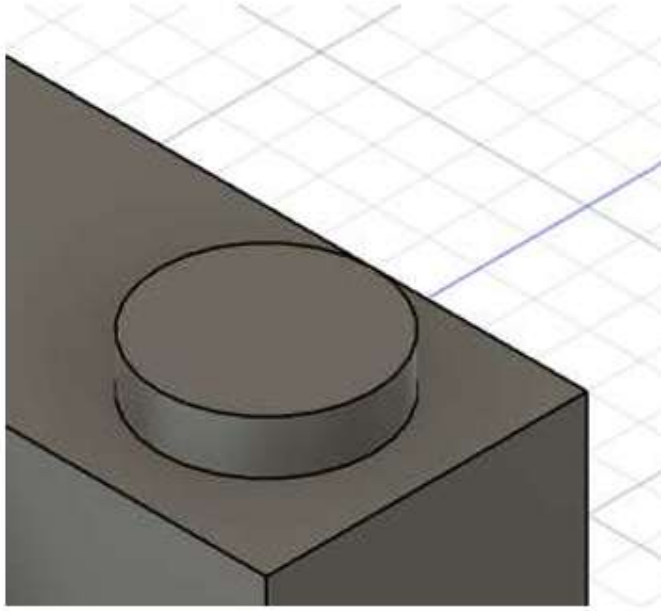
1. Create Sketch: Click Create Sketch and select the Top Face of the main camera body.
2. Draw Circle:
    - Use the Center Diameter Circle tool (C).
    - Position the circle using sketch dimensions: 15 mm from the right edge and 15 mm from the back edge (referencing the layout dimensions).
    - Set the diameter to 20 mm.



3. Finish Sketch: Click Finish Sketch.

4. Extrude 4 Details:
    - Press E and select the circular profile.
    - Distance: 5 mm (height of the button protrusion).
    - Operation: Join.
    - Click OK.

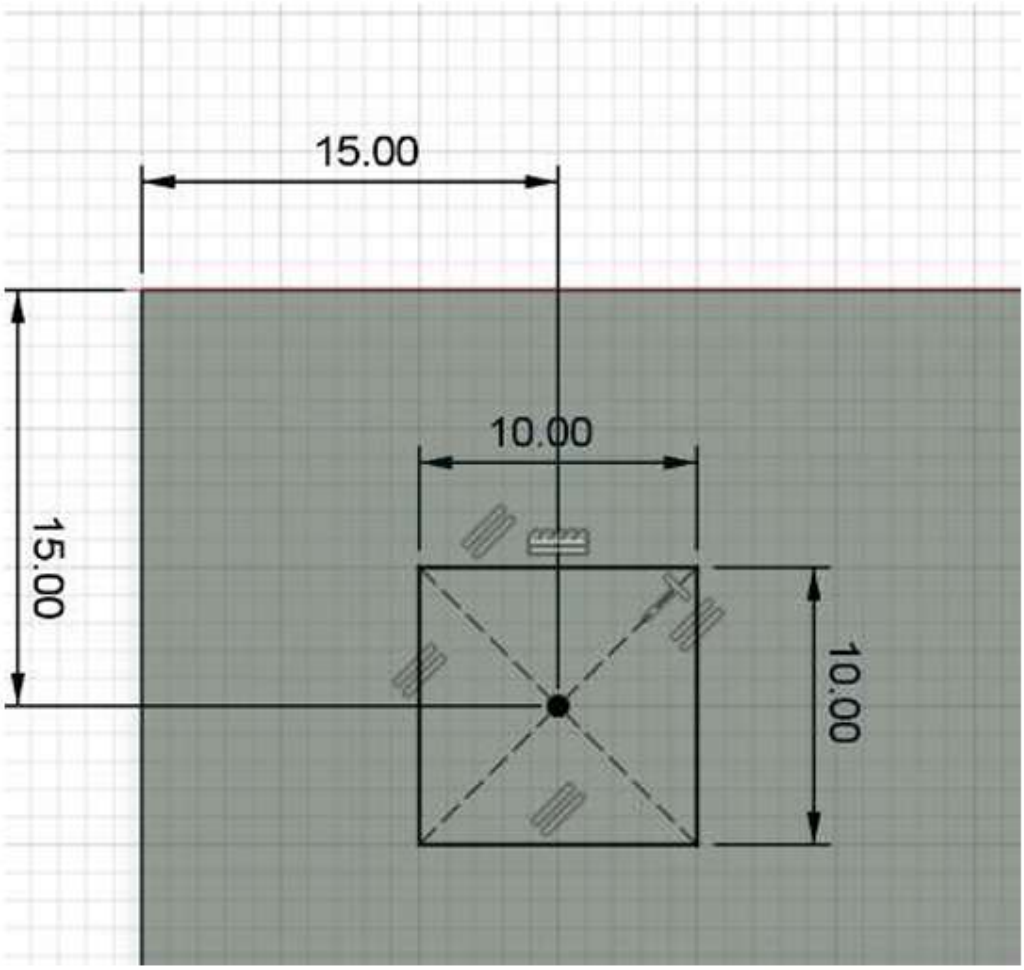




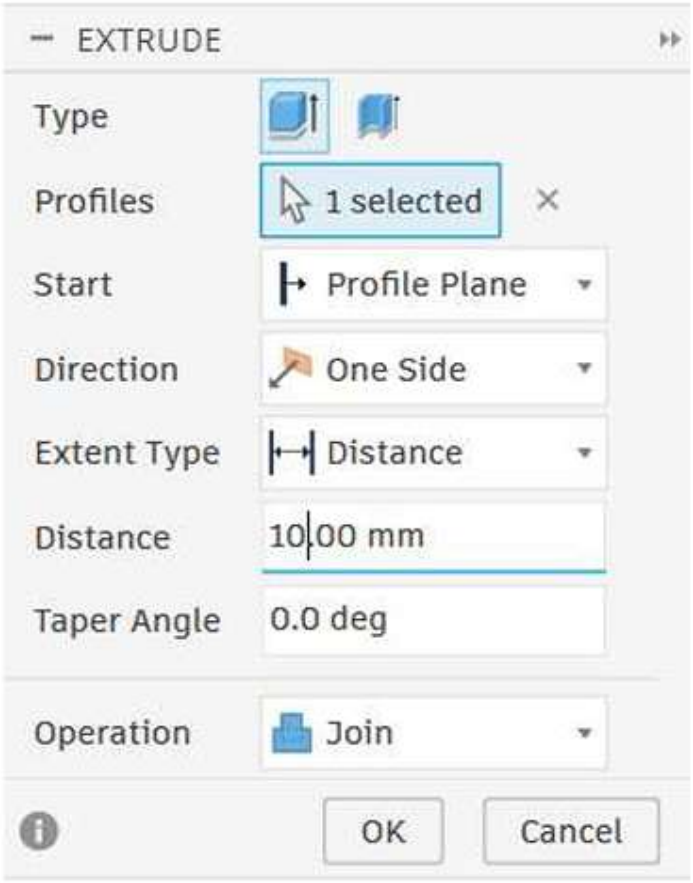
Step 4: Creating the Top-Left Square Button

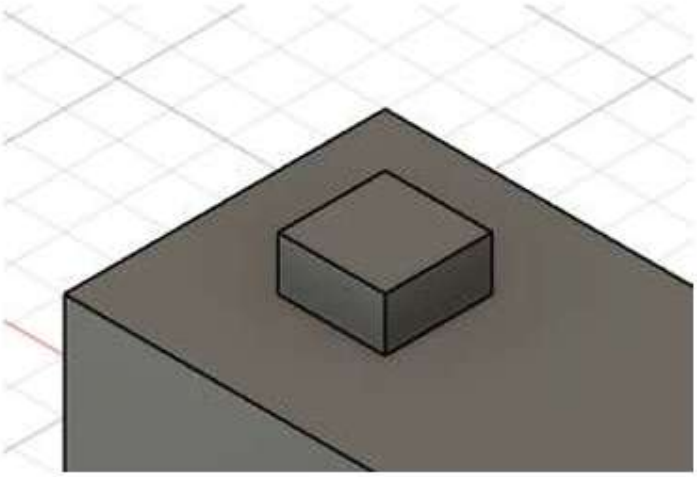
Objective: Model the square-shaped functional button/flash housing on the top left of the camera body.

1. Create Sketch: Click Create Sketch and select the Top Face of the main camera body.
2. Draw Square:
    - Use the Center Rectangle or Two-Point Rectangle tool.
    - Position it with dimensions: 15 mm from the left edge and 15 mm from the back edge.
    - Set the dimensions of the square to 15 mm × 15 mm.



3. Finish Sketch: Click Finish Sketch.
4. Extrude 5 Details:
    - Press E and select the square profile.
    - Distance: 5 mm (height of the square button).
    - Operation: Join.
    - Click OK.



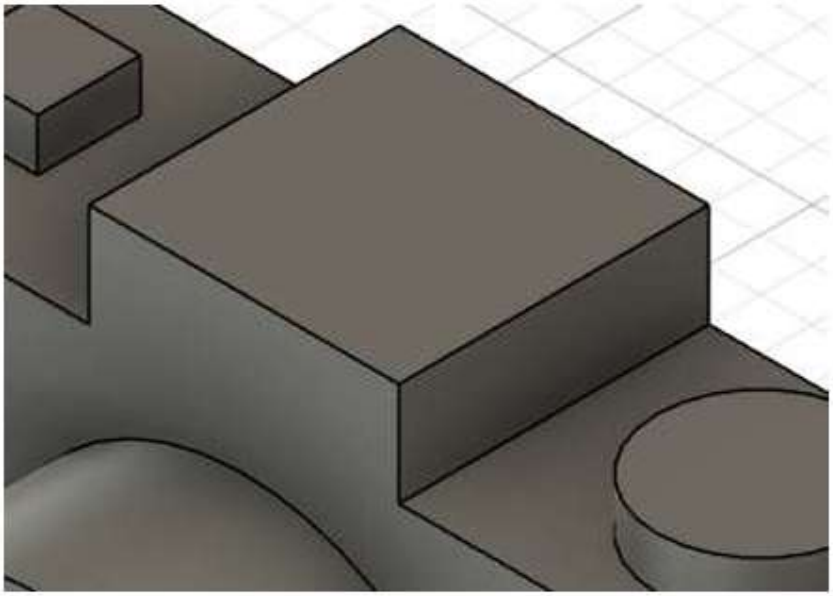


Step 5: Creating the Top Central Prism / Viewfinder Housing (Extrude & Chamfer)

Objective: Model the raised central housing block on top of the camera and shape its front profile with a 45° bevel.

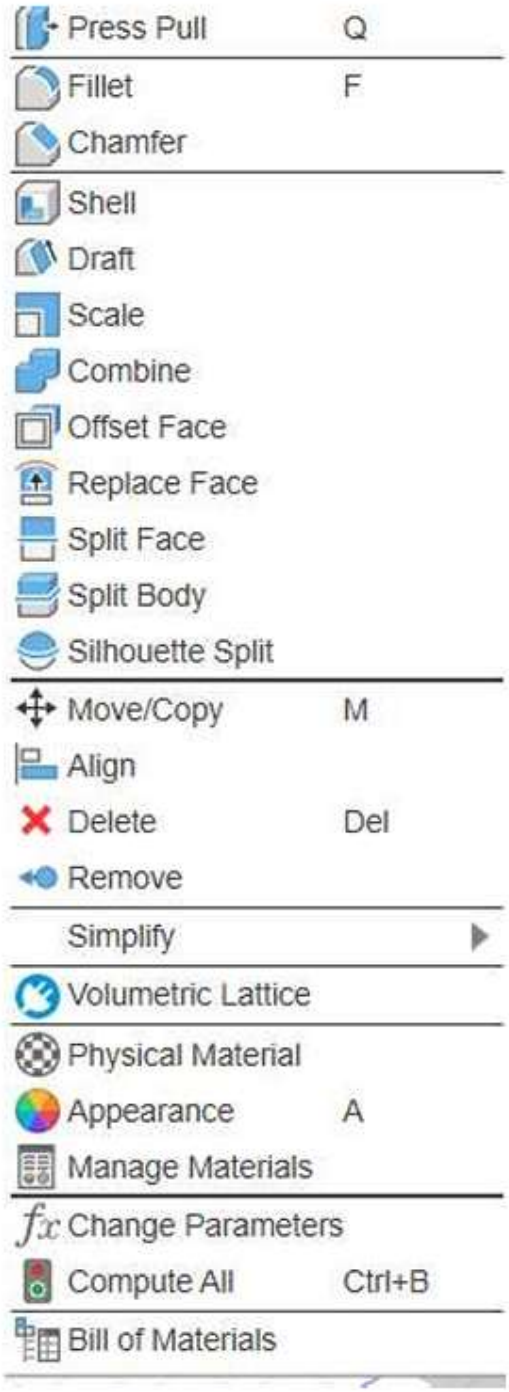
1. Sketching and Extruding the Housing Block

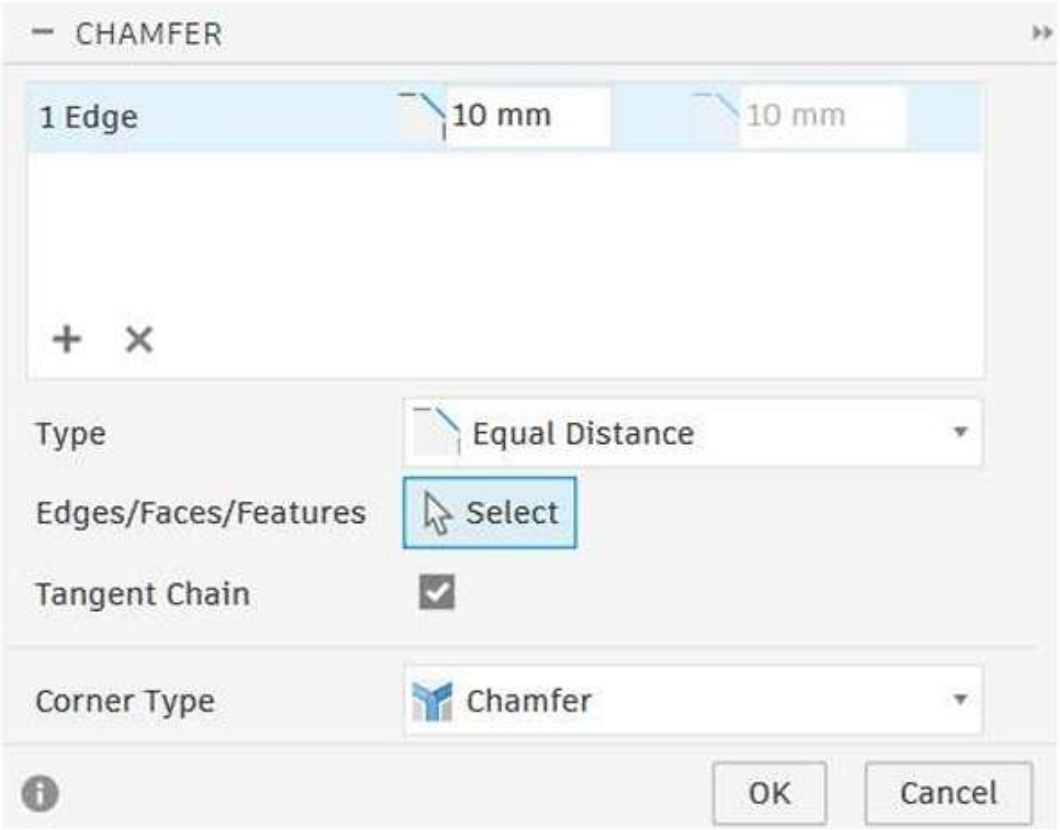
- Select Sketch Plane: Click Create Sketch in the top toolbar and select the Top Face of the main camera body.
- Draw Profile: Select the Two-Point Rectangle tool, draw the footprint on the top face, and apply dimensions based on the drawing:
    - Width: 30 mm
    - Depth/Length: 30 mm
    - Position: Offset 10 mm forward from the rear edge of the camera body and horizontally centered.
- Finish Sketch: Click Finish Sketch.
- Extrude: Press E, select the rectangular profile, set the Direction to One-side, enter a Distance of 10 mm (housing height), set the Operation to Join, and click OK.

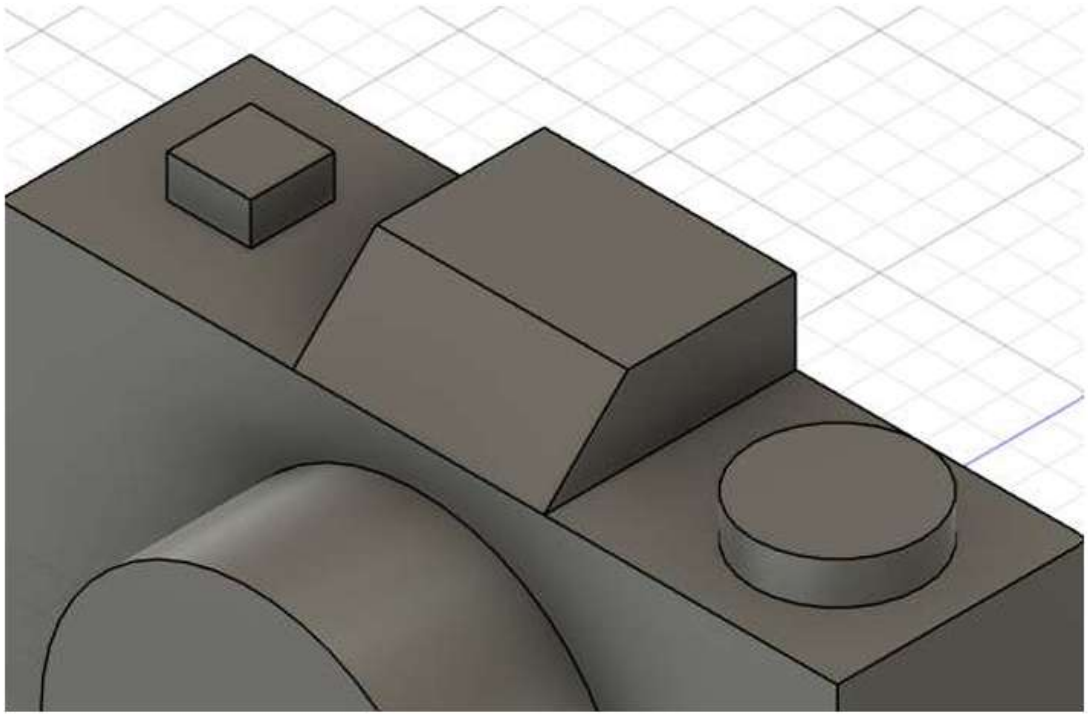


2. Applying the 45° Chamfer

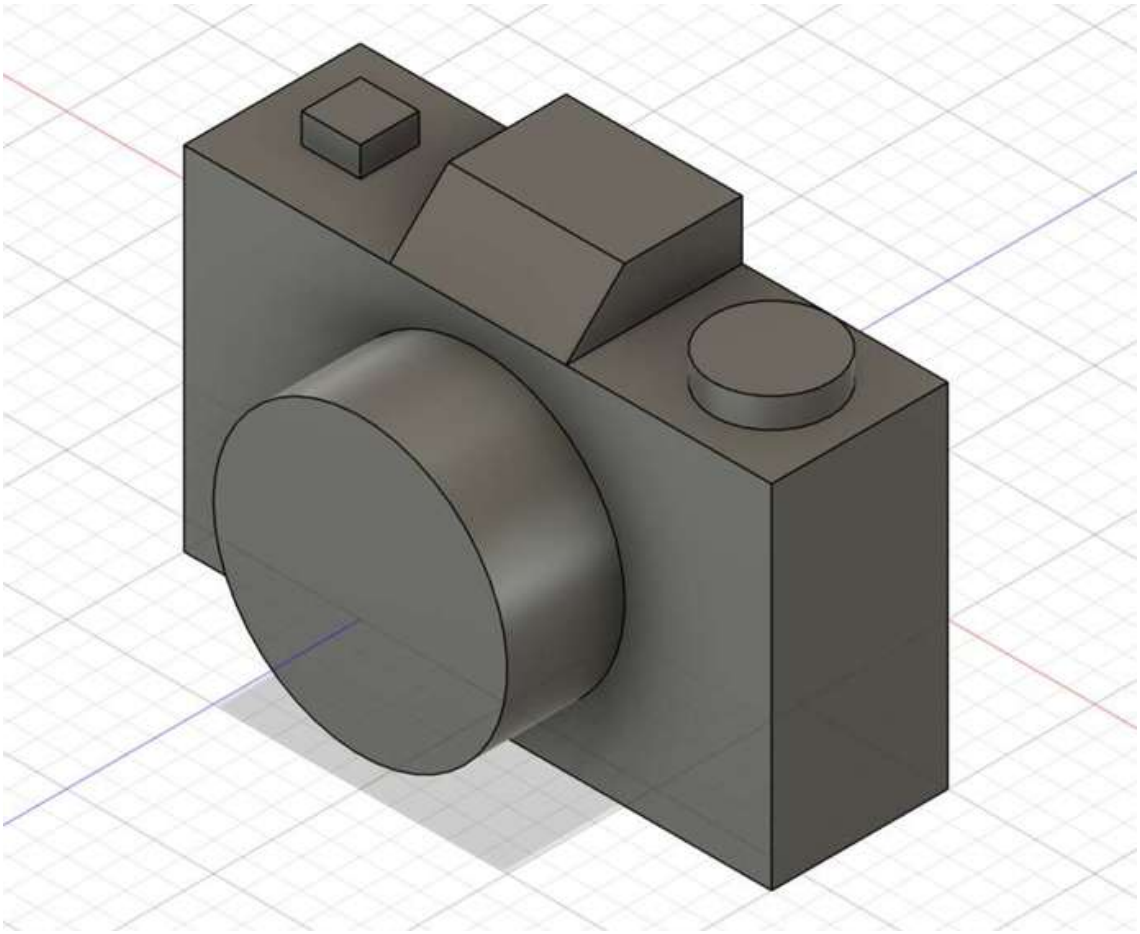
- Access Tool: Go to the top menu and select Modify > Chamfer.
- Select Edge: Click on the top-front horizontal edge of the newly extruded housing block.
- Configure Settings: Set the chamfer type to Distance and Angle, and enter an angle of 45° to match the drawing's specified bevel.
- Finalize: Review the preview to ensure the slanted prism shape is correct, then click OK.

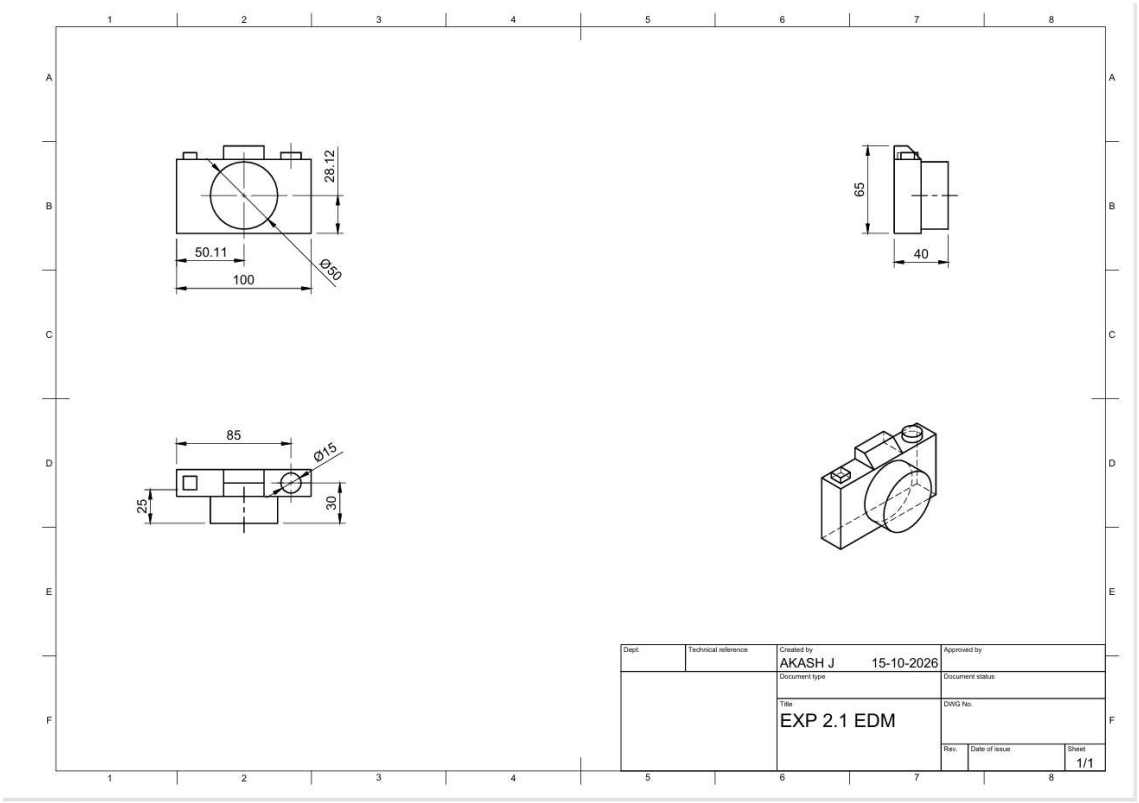

Press Pull Q
Fillet F
Chamfer
Shell
Draft
Scale
Combine
Offset Face
Replace Face
Split Face
Split Body
Silhouette Split
Move/Copy M
Align
Delete Del
Remove
Simplify
Volumetric Lattice
Physical Material
Appearance A
Manage Materials
Change Parameters
Compute All Ctrl+B
Bill of Materials


CHAMFER
1 Edge
10 mm
10 mm
Type
Equal Distance
Edges/Faces/Features
Select
Tangent Chain
Corner Type
Chamfer
OK
Cancel



OUTPUT





RESULT

Thus, the 3D model is created from the given isometric drawing.